Jacob Daniel Moss

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Academic

University of Cambridge, Computer Lab 2020 - 2024

PhD in Machine Learning

Researched mechanistic inductive biases in machine learning models. Improved a popular biological dynamical model, RNA velocity, by removing unrealistic assumptions and providing a data-driven approach to integrate single cell epigenetic markers and continuous transcription rates. This was achieved by designing a transformer-based diffusion model to generate trajectories of cell fate with complex system dynamics.

Advisers: Prof. Pietro Liò & Prof. Jeremy England.

Research areas: machine learning in single cell genetics, dynamical modelling.

University of Cambridge, Computer Lab 2019 - 2020

MPhil Advanced Computer Science: **Distinction**

Developing machine learning models for the stochastic modelling of dynamics. Extended the Neural ODE framework to a distribution of initial conditions. Investigated methods of solving PDE-based latent force models.

Adviser: Prof. Pietro Liò.

Research areas: machine learning, drug discovery, genomics.

Specialism: Probabilistic Machine Learning, Digital Signal Processing, Natural Language Processing

University College London 2016 - 2019

BSc Computer Science: **First class honours**, top 10% of cohort

Final project on large-scale data aggregation of unstructured auction house data, with a grammar-based NLP processing pipeline. Courses in Statistics, Machine Learning, Linear Algebra, Stochastic Calculus.

Adviser: Prof. Philip Treleaven

Thesis: "Cross-Auction Arbitrage with Machine Learning"

Experience

Machine Learning Engineer – Biographica Sep 2024 - current

Building large genomic models for gene edit discovery, ranking and trait identification. Engineering foundation models and large CNN-transformer hybrids to work on complex and noisy biological sequence data. Built scalable entity recognition and data orchestration for processing large datasets with inconsistent metadata.

Freelance AI Contracting Feb 2023 - Sep 2024

- **Trading & Game theory**: Developed a low-latency crypto trading bot initiating orders on several global cryptocurrency exchanges. Deployed quantised GPT model using AWS Neuron to tag news events. For the same client, implemented a Pluribus-style poker bot capable of superhuman performance on 2p no-limit Texas hold 'em using counter-factual regret minimisation and Monte Carlo tree search.
- **LandSpec**: Built a real estate application and microservices infrastructure to serve AI chat, summarisation, and semantic search alongside vector DBs and scrapers.
- **London Stock Exchange Group**: Led a research project investigating uses of AI within indexing platforms.
- **Myria Biosciences:** Creation of a graph-based bioinformatics platform and large sequence model training for antibiotic discovery. Integrated a novel masked objective in the protein domain.
- **Aviva:** Ran workshops on LLMs within the risk, investments, and insurance industry to senior executives.

Research Secondment – GlaxoSmithKline Jun 2022 - Sep 2022

Worked alongside Ed Curry in functional genomics on the analysis of a high-throughput CRISPR assay, focussing on determining genetic interactions and drug targets.

Machine Learning Engineer – Synteny Jan 2022 - Jun 2022

Developed machine learning methods for protein embedding and disease-sequence association prediction. Implemented several models from literature, extending to larger datasets and evaluating performance on a large range of experimental datasets.

Machine Learning Engineer – Cassandre Investments Nov 2018 - May 2019

Built an asset price prediction model using autoregressive Gaussian processes for forecasting. Feature engineering including grammar-based named entity extraction and sale frequency calculations. NLP for clustering items based on description.

Data Science Internship – Ocado Technology Jul 2018 - Sep 2019

Developed an ensemble classifier to determine the sentiment of Ocado-related tweets at 80% accuracy. Wrote an algorithm to incorporate a new sentiment class in pre-trained models.

Projects:

- **Ticketing Platform:** executed £270k+ in sales with secure QR entry and phased releases.
- **UCL:** led a research group at UCL investigating machine-executable natural language legal contracts.

Skills

Machine learning: PyTorch, TensorFlow, numpy, pandas

Programming languages: Python, C++, Java, C#, JavaScript

Frameworks: AWS (incl. Neuron), GCP, Dagster, React, RxJS, .NET, Django, Kubernetes, git, SQL

Publications

University of Cambridge, AI Research Group
Dec 2019 - Nov 2024

- **Moss, J. D.**, England, J. L., and Liò, P., 2023. Pseudotime Diffusion. *NeurIPS 2023 Machine Learning and the Physical Sciences workshop.*
  Analysis of whole-genome sequencing data has been outpaced by the experimental techniques that generate those datasets. Conventional methods like dimensionality reduction limit the information obtainable from a dataset. We expand on the biophysical model of RNA velocity to yield meaningful insights into the functional trajectories of individual cells, leading to downstream applications such as the identification of genes driving a disease pathway. We achieve this by learning a diffusion model to compute pseudotime and velocity distributions.
- **Moss, J. D.**, Opolka, F. L., England, J. L., and Liò, P., 2023. Deep Kernel Learning of Nonlinear Dynamical Systems. *TMLR* (also *ICML 2023 SynS & ML workshop*).
  Scientific processes are often modelled by sets of differential equations. We drastically improving the scalability of latent force models, balancing data-driven and mechanistic inference in dynamical systems, whilst improving inference accuracy. This overcomes the computational challenges associated with quantifying uncertainty in large datasets. We learn the solution operator by employing a deep kernel along with a function embedding.
- **Moss, J. D.**, Opolka, F. L., Dumitrascu, D., and Liò, P., 2021. Gene Regulatory Network Inference with Latent Force Models. *NeurIPS 2021 Machine Learning and the Physical Sciences workshop & AAAI 2021 Science-Guided AI symposium.*
  The existing inference techniques associated with physically inspired latent force models rely on the exact computation of posterior kernel terms which are seldom available in analytical form, rendering them intractable to most applications. We overcome these problems by proposing a variational solution to a general class of non-linear and parabolic partial differential equation latent force models. Further, we show that a neural operator approach can scale our model to thousands of instances, enabling fast, distributed computation. We achieve competitive performance on several tasks where the kernels are of varying degrees of tractability.
- Norcliffe, A., Bodnar, C., Day, B., **Moss, J. D.**, & Liò, P., 2021. Neural ODE Processes. In *ICLR 2021 main conference* and *NeurIPS 2020 Machine Learning and the Physical Sciences workshop*
  We introduce Neural ODE Processes (NDPs), a new class of stochastic processes determined by a distribution over Neural Ordinary Differential Equations (NODEs). This resolves two issues with NODEs: the inability to adapt to incoming data-points and the inability to explain a sparse set of measurements with many possible underlying dynamics. We show that our model can successfully capture the dynamics of low-dimensional systems from just a few data-points. At the same time, we demonstrate that NDPs scale up to challenging high-dimensional time-series with unknown latent dynamics such as rotating MNIST digits.
- Norcliffe, A., Day, B., **Moss, J. D.**, & Lió, P., 2021. Meta-learning using privileged information for dynamics. In *ICLR 2021 Learning to Learn and SimDL workshops.*

Achievements

GSK.ai PhD Award 2024

Stanford Machine Learning Course
Stanford Coursera courses on Sequence Models and Deep learning. Certs: GUF2CZ62FQHD, 8US78QT64Q3K

Grants & Exhibitions
Accelerate C2D3 Grant 2023: Awarded the Accelerate-C2D3 grant for AI research and innovation.
Wearable Technology Show 2016: Exhibited the Visijax jacket: a smart cyclist jacket with fabric-integrated LEDs and inbuilt MCU controller providing navigational prompting and geofencing.